VARSHA VIKRAMAN

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PROFILE

- Full-Stack Developer with hands-on professional experience building production MERN Stack (MongoDB, Express.js, React.js, Node.js) web applications, secure REST (Representational State Transfer) APIs with JSON Web Token (JWT) authentication and Role-Based Access Control (RBAC).
- Hands-on experience integrating Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), multilingual AI, vector databases, and AI assistants for enterprise applications.
- Experienced with containerized deployment using Docker and Kubernetes (K8s), including managing GPU-accelerated AI inference workloads on AWS cloud
- Deep interest in Blockchain and Web3 fundamentals, including smart contract interaction and decentralized architecture, backed by a PG Diploma in Blockchain Technology.

EXPERIENCE

Full Stack Developer Mar 2026 – Present
ZenturioTech — Thiruvananthapuram, Kerala

- Develop and maintain full-stack web applications using React.js, Node.js, Express.js, and MongoDB, contributing to both frontend UI and backend API layers.
- Designed JWT + role-based access control to ensure secure data handling.
- Delivered responsive interfaces with Tailwind CSS; reduced cross-device layout issues through systematic component audit.
- Deployed and managed Kubernetes-based production environments running AI inference workloads (ASR, TTS, multimodal LLM pods)
- Containerized services with Docker, enabling consistent dev/staging/prod environments across the team.
- **Telehealth Platform (Confidential Client)** *React.js, TypeScript, Node.js, Express.js, MongoDB, Jitsi*
  - Built core features for a telehealth platform connecting clients with clinicians, including a therapist dashboard with session scheduling and calendar management.
  - Implemented JWT-authenticated video calling via Jitsi, and chat functionality including a message composer with emoji support, file uploads, and voice message recording/playback.
  - Resolved cross-browser CORS issues affecting voice message playback and fixed UI bugs across profile views, calendar widgets, and session cards.
- **Mindfulness Music Game (Confidential Client)** *React.js, PixiJS, Tone.js*
  - Built a mindfulness-focused, music-driven bubble-tapping game featuring a custom animated mascot character.
  - Implemented a Guided Melody Mode with live target-note derivation and musical staff notation rendering.
  - Built canvas-based sprite sheet animation (multi-row, ping-pong playback) and audio export functionality via a CDN-based encoding workaround.
  - Diagnosed and fixed stale-closure bugs in the game's render/tick loop; polished UI across game states and mobile layouts.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3, Python
Frontend: React.js, React Native, Tailwind CSS
Backend: Node.js, Express.js, NestJS, REST APIs
Database: MongoDB, SQLite
DevOps: Docker, Kubernetes, Github
Tools: Postman, Figma
Blockchain: Solidity, Ethereum, Hardhat, Web3.js, viem, Hyperledger Fabric
AI Technologies: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, Prompt Engineering, Hugging Face, vLLM, Groq API

PROJECTS

AI Government Assistant GitHub
Technologies: React.js, Node.js, Express.js, MongoDB, MongoDB Atlas Vector Search, LangChain, Groq, HuggingFace Embeddings

- Built an AI-powered Human Capital Intelligence platform for Kerala Local Self Government employees, combining a document-grounded AI assistant with employee and departmental data management.
- Designed and implemented a full RAG pipeline (LangChain, Groq, MongoDB Atlas Vector Search, HuggingFace embeddings) with PDF ingestion, per-page citations, and department/district-based filtering of government circulars and news.
- Built an AI-based department classifier (Groq) to auto-route and tag incoming documents, and developed CSV employee roster and bulk PDF import tools to streamline onboarding of institutional data.
- Led a bilingual Malayalam/English upgrade for the AI assistant, addressing i18next/AI-content language synchronization and evaluating multilingual embedding models for regional-language support.
- Debugged and resolved schema divergences across the platform (user model field mismatches, role casing, enum constraints) and migrated data-fetching logic to React Query v5.

Live Demo

Personal Finance Advisor GitHub

Technologies: React.js, Tailwind CSS, Node.js, Express.js, MongoDB, Docker

- Full-stack budgeting app with income/expense tracking, savings goals, and visual analytics dashboards.
- Automated monthly summaries with goal-progress indicators; containerized for one-command local setup via Docker.

Blog App (Mobile) GitHub

Technologies: React Native (Expo), NestJS, TypeORM, SQLite

- Cross-platform mobile app with full CRUD, offline draft storage via AsyncStorage, and push notifications.
- SQLite + TypeORM for structured local persistence; NestJS backend for clean, modular API design.

Decentralized Energy Trading DApp GitHub

Technologies: React.js, Solidity, Hardhat, viem, Tailwind CSS, Ethereum

- Peer-to-peer energy marketplace connecting renewable producers and consumers via on-chain Energy Tokens (ET), each representing 1 kWh of verified output.
- Smart contracts provide trustless, instant settlement with full on-chain transaction history — no intermediary.

EDUCATION

PG Diploma in Blockchain Technology 2025
Digital University Kerala, Thiruvananthapuram

B.Tech in Electronics and Communication Engineering 2023
Government Engineering College, Kozhikode